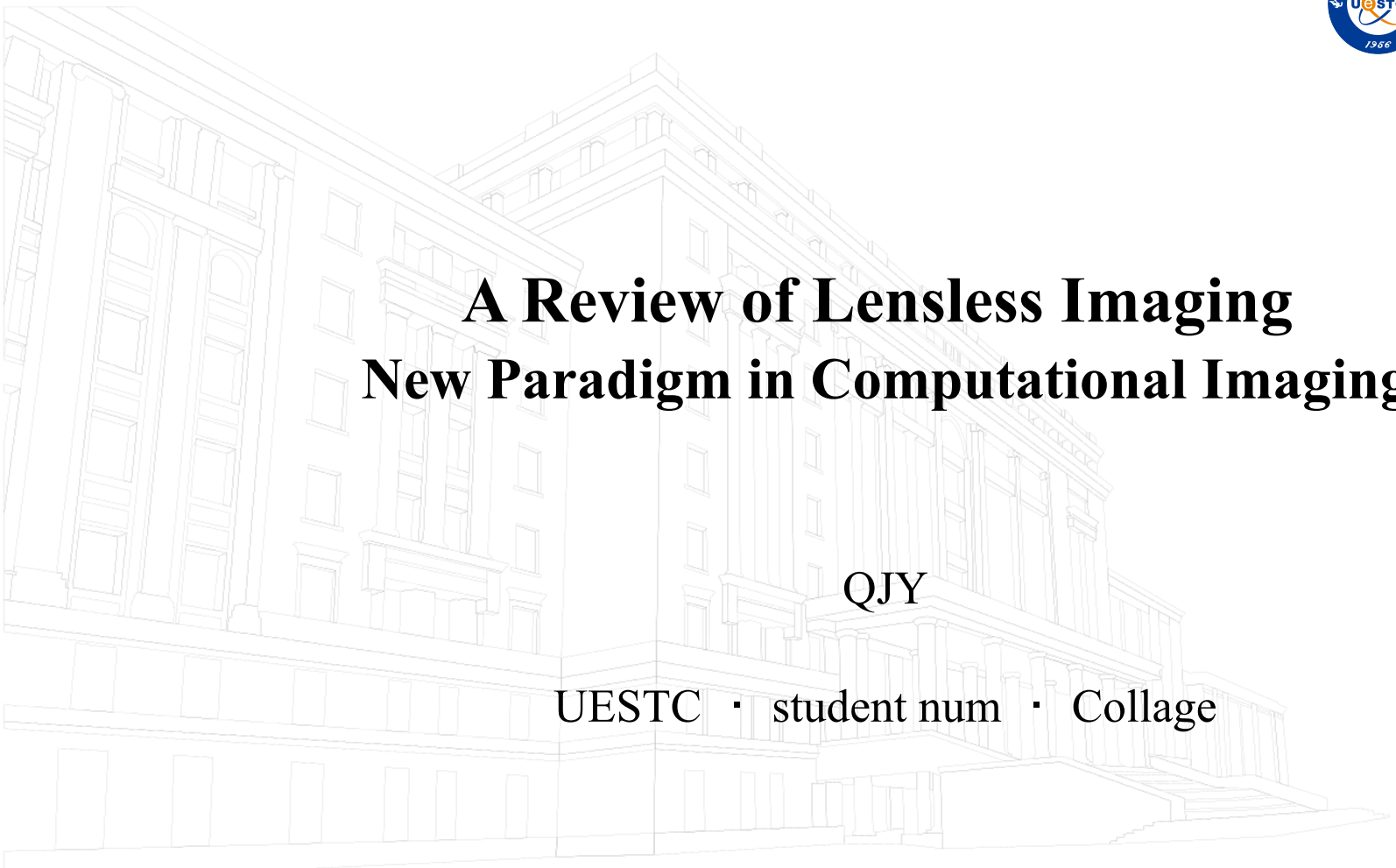


A Review of Lensless Imaging
New Paradigm in Computational Imaging

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Outline

What is lensless imaging?

- One-sentence introduction
- Lens-based Camera vs. Lensless Camera
- Applications of Lensless Cameras

How does a lensless camera work?

- System Architecture
- Hardware (Encoding Optics)
- Software (Decoding algorithm)

What am I working on?

- Relevance to My Work ...
- References

What is lensless imaging?

What is lensless imaging?

One-sentence introduction

A lensless camera is a camera without a lens!

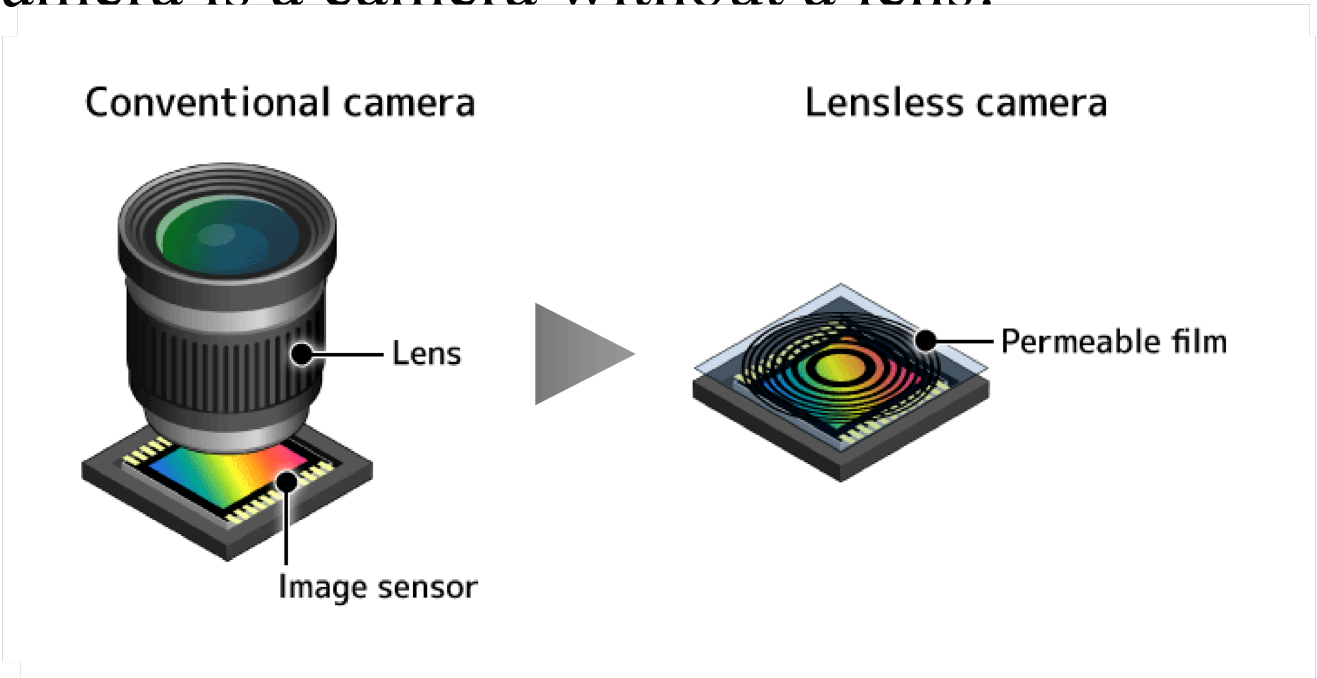


图 1　This image shows the appearance of a lensless camera.[1]

What is lensless imaging?

Lens-based Camera vs. Lensless Camera

Aspect	Lens-based	Lensless
Imaging Principle	Lens focus	Mask + algorithm
Size	Thick & heavy	Thin & light
Cost	High	Low
Integration on-chip	Impossible	Possible
Image Quality	Depends on lens	Depends on reconstruction algorithm
High-dimensional Imaging	Multiple shots	One shot

What is lensless imaging?

Applications of Lensless Cameras

Microscopic imaging

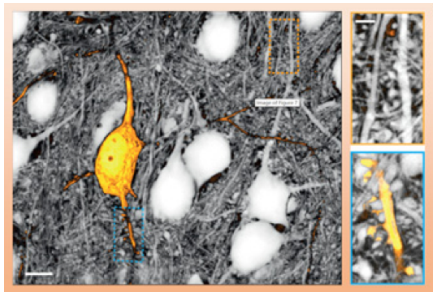


图 2　Brain tissue imaging via SUSHI microscopy[2]

3D imaging

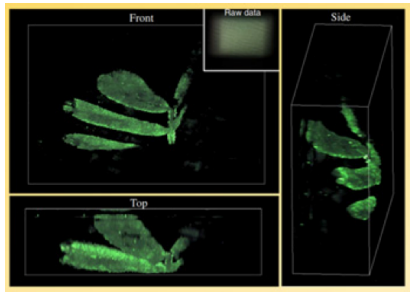


图 4　Single-shot 3D imaging using DiffuserCam[4].

Multispectral imaging

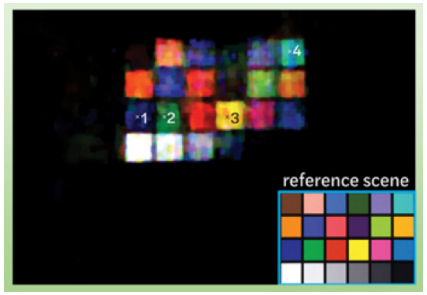


图 3　Snapshot hyperspectral imaging with Spectral DiffuserCam[3].

Optical fiber imaging

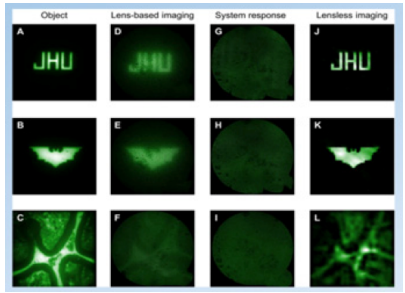


图 5　Lensless microendoscopy via masked optical fiber[5].

How does a lensless camera work?

How does a lensless camera work?

System Architecture

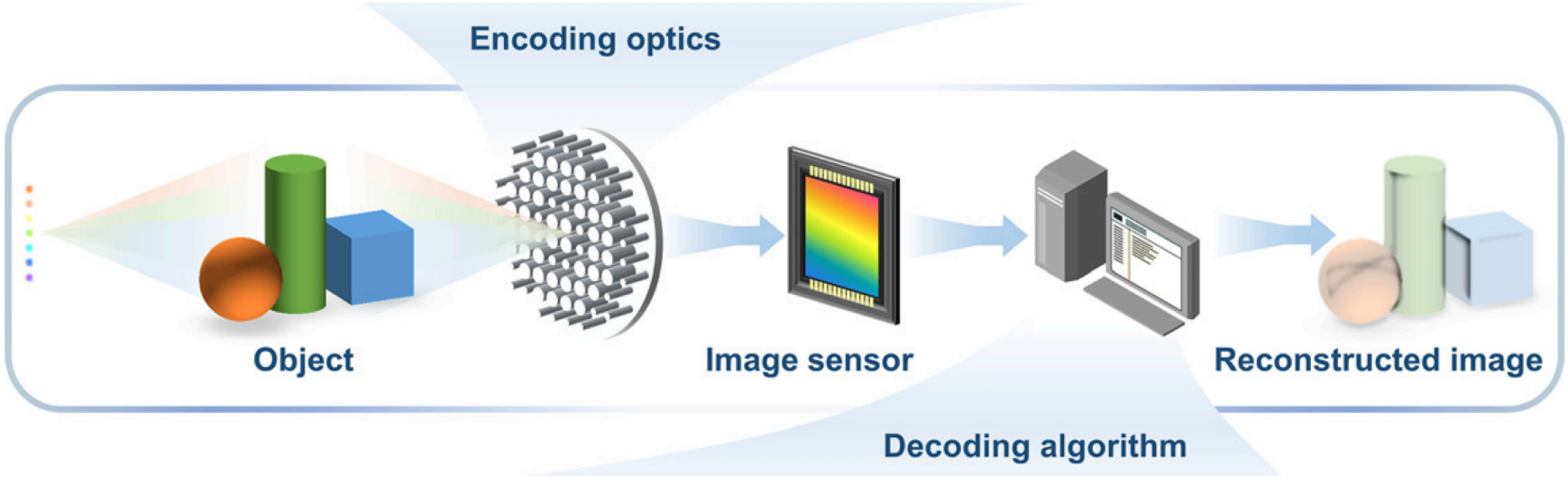


图 6　Overview of a typical lensless imaging system[1].

How does a lensless camera work?

Hardware (Encoding Optics)

Introduction

- Inspired: pinhole imaging
- Feature: forms multiplexed patterns

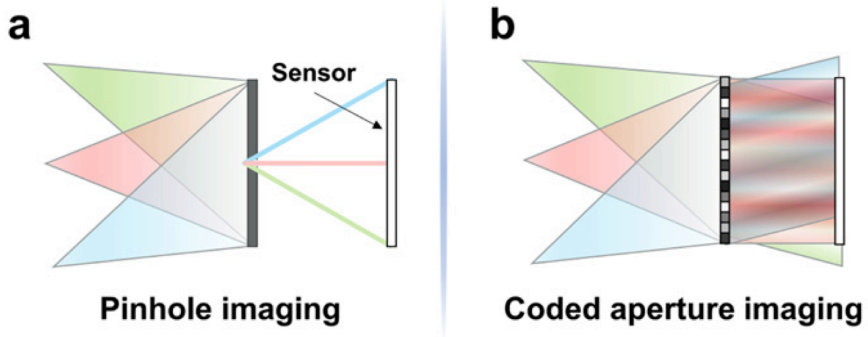
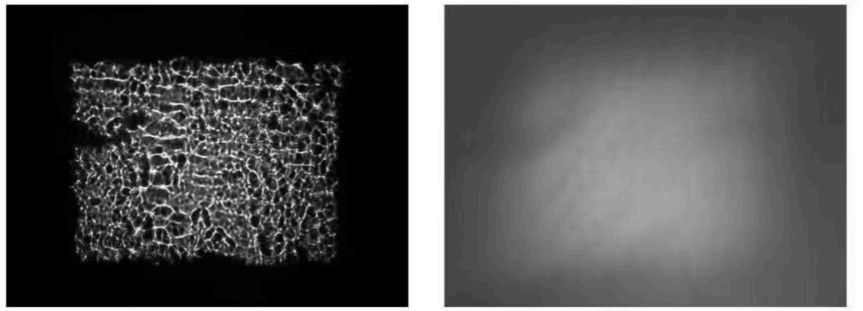


图 7　Comparison between pinhole and coded aperture imaging[1].

Example

- Fig a: projection of the mask
- Fig b: captured multiplexed measurements



(a) Caustic image of a single point source　(b) Sensor reading of a hand

图 8　Examples of mask projection and sensor measurement[6].

How does a lensless camera work?

Software (Decoding algorithm)

Problem Description

- Forward model:

$$y = Ax + n \tag{1}$$

  - A: an underdetermined matrix
  - n: noise
- Goal: Solve for x

Method

- Optimization method:

$$\min_x \| Ax - y \|^2 + R(x) \tag{2}$$

  - Iterative solvers: ISTA, ADMM, FISTA
- Deep Learning

$$x = f_\theta(y) \tag{3}$$

  - Models: UNet, Unfold network, Deep noise prior

Illustration diagram

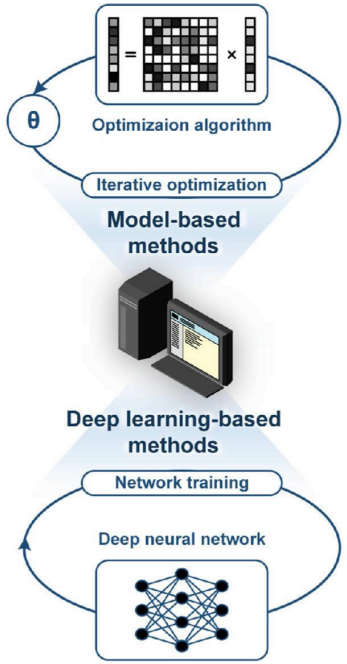


图 9 iterative optimization & deep learning-based reconstruction approaches[1].

What am I working on?

What am I working on?

Relevance to My Work ...

- Focus on lensless image reconstruction.
- Now:
  - Using new models.
  - Improve robustness to noise.
  - Build a complete system.

References

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[3] K. Monakhova, K. Yanny, N. Aggarwal, 和 L. Waller, 《Spectral DiffuserCam: lensless snapshot hyperspectral imaging with a spectral filter array》, *Optica*, 卷 7, 期 10, 页 1298 – 1307, 2020.

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[6] G. Kuo, N. Antipa, R. Ng, 和 L. Waller, 《DiffuserCam: diffuser-based lensless cameras》, 收入 *Computational Optical Sensing and Imaging*, 2017, 页 CTu3B – 2.

End of Beamer!